AI & Social Media Impact on Student Health

A Python-based Exploratory Data Analysis Project

Document Overview

This document explains the project as implemented in the supplied notebook. It covers the project objective, dataset, tools, data inspection and cleaning, derived variables, aggregation, exploratory visualizations, descriptive results, limitations, and recommendations for completing or extending the analysis.

- Executive Summary
- Problem Statement and Objectives
- Dataset Description
- Tools and Environment
- Workflow and Data Preparation
- Derived Features and Categories
- Aggregation and Exploratory Data Analysis
- Key Descriptive Results
- Interpretation of the Visualizations

Executive Summary

The project, titled "AI & Social Media Impact on Student Health," examines student digital habits and well-being using a tabular dataset. The notebook uses Python with NumPy, pandas, Matplotlib, and Seaborn to load the CSV, inspect its structure, check for duplicate identifiers and missing values, adjust categorical data types, create additional analysis fields, aggregate selected measures, and generate exploratory charts.

Measure	Dataset result
Student records	16,000
Original dataset columns	10
Average age	19.04 years
Average social-media use	4.54 hours/day
Average AI-tool use	2.59 hours/day
Average sleep	6.55 hours/day
Average physical activity	1.25 hours/day
Average mental-health score	72.49
Average physical-health score	88.02

The dataset contains 16,000 records and 10 original columns. In the supplied data, the average daily social-media use is about 4.54 hours and AI-tool use is about 2.59 hours. The average sleep duration is about 6.55 hours. These figures summarize the dataset and should not be treated as proof that digital usage causes a particular health outcome.

Problem Statement and Objectives

The notebook frames the problem around students spending substantial time on social media and AI tools, while the relationship between those habits, sleep, physical activity, and mental and physical health is not immediately clear. The intended audience includes educators and parents seeking data-driven insight into student digital behavior and well-being.

Project objectives

- Load and understand the student lifestyle dataset.
- Inspect row count, column names, data types, sample records, duplicate student IDs, and missing values.
- Create useful derived variables such as combined screen time, overall health score, age group, health category, sleep category, and AI-use group.
- Summarize usage and health measures by education level and gender.
- Use visualizations to explore distributions, demographic composition, and the relationship between social-media use and mental-health scores.
- Provide an evidence-based foundation for further analysis and student digital-wellbeing discussions.

Domain and intended outcome

Domain: Education, with a focus on student well-being and digital behavior. The notebook's stated outcome is to uncover patterns and trends that can support informed decisions. The implemented workflow is exploratory rather than causal or predictive.

Dataset Description

The notebook identifies the dataset source as a Kaggle/Open Data Portal dataset and states a collection timeline of 2025–2026. The CSV loaded in the notebook is named AI_SocialMedia_Student_Dataset.csv. The original table has 16,000 rows and 10 columns.

Source listed in the notebook: Kaggle, "AI and Social Media Impact: Student Health and Grades." The exact provenance and collection methodology are not independently verified in this document.

Field	Meaning in the notebook	Type / role
Student_ID	Unique identifier for each student record	Identifier
Age	Student age in years; described as 13–25	Numeric / demographic
Gender	Male, Female, or Non-binary	Categorical
Education_Level	High School, College, or University	Categorical
Daily_Social_Media_Hours	Average social-media hours per day	Numeric / usage
Daily_AI_Tool_Usage_Hours	Average AI-tool usage hours per day	Numeric / usage
Sleep_Hours	Average sleep hours per day	Numeric / lifestyle
Physical_Activity_Hours	Average physical-activity hours per day	Numeric / lifestyle
Mental_Health_Score	Mental well-being score; higher is described as better	Numeric / outcome
Physical_Health_Score	Physical well-being score; higher is described as better	Numeric / outcome

Tools and Environment

Tool / library	Purpose in this project
Google Colab / Jupyter notebook	Interactive environment for running Python cells and displaying tables and plots.
Python	Programming language used for analysis.
NumPy (numpy)	Numerical computing support; imported in the notebook.
pandas	CSV loading, data-frame inspection, type conversion, missing-value checks, binning, grouping, and summaries.
Matplotlib (pyplot)	Figure and axis creation, chart labels, titles, and display.

Seaborn	Statistical visualizations including histograms, count plots, box plots, and scatter plots.

Workflow and Data Preparation

Import libraries: Imports NumPy, pandas, Matplotlib pyplot, and Seaborn.

Load the CSV: Reads AI_SocialMedia_Student_Dataset.csv into the data frame named df and prints a success message.

Inspect the dataset: Displays the full data frame, shape, row and column counts, first and last rows, column names, info, and data types.

Check identifiers: Counts duplicated values in Student_ID; the saved notebook output reports 0 duplicated IDs.

Check missingness: Uses df.isnull() and df.isnull().sum(); saved output reports zero missing values in each of the 10 original columns.

Set categorical types: Converts Gender and Education_Level to pandas category data types.

Create analysis fields: Computes total screen time and overall health score, then adds age, health, sleep, and AI-use categories.

Aggregate data: Groups social-media use and mental-health scores by education level; summarizes mental and physical scores by gender using mean and median.

Explore and visualize: Runs descriptive statistics and creates six core charts with explanatory markdown.

Data-quality checks

Check	Notebook operation	Observed result / note
Dataset dimensions	`df.shape`	16,000 rows × 10 columns.
Duplicate identifiers	`df['Student_ID'].duplicated().sum()`	Saved output: 0 duplicate IDs.
Missing values	`df.isnull().sum()`	Saved output: 0 missing values in every original column.
Data types	`df.info()` and `df.dtypes`	Numeric columns remain numeric; Gender and Education_Level are explicitly converted to category.

The notebook performs checks but does not show a separate imputation, outlier-removal, or row-removal stage. Because no missing values were reported, the notebook continues with the available records.

Derived Features and Categories

The notebook creates the following analysis fields or calculated values:

Field / expression	Definition	Purpose / caution
Total_Screen_Time	Daily_Social_Media_Hours + Daily_AI_Tool_Usage_Hours	Combined daily usage indicator. In the notebook this is assigned to a standalone Series, not added as a column to df.
Overall_Health_Score	(Mental_Health_Score + Physical_Health_Score) / 2	Simple unweighted average of the two health scores; assumes the scales are comparable.
Age_Group	13–15, 16–18, 19–21, 22–25	Groups age for count plots and demographic comparisons.
Health_Category	At Risk: 0–60; Average: 60–75; Healthy: 75–100, based on Overall_Health_Score	Threshold categories created with pandas cut; exact boundary handling follows pd.cut's default right-closed intervals.
Sleep_Category	Short (<6h), Normal (6–8h), Long (>8h)	Groups sleep duration. Actual interval edges are defined by bins [0, 6, 8, 24].
AI_Usage_Group	Low (0–1h), Moderate (1–3h), High (3–5h), Very High (5h+)	Groups daily AI-tool usage using bins [-0.1, 1, 3, 5, 24].

Aggregation and Exploratory Data Analysis

Grouped summaries

- Education-level summary: calculates mean Daily_Social_Media_Hours and Mental_Health_Score by Education_Level, rounded to two decimals.
- Gender summary: calculates mean and median Mental_Health_Score and Physical_Health_Score by Gender, rounded to two decimals.
- Education-level box-plot summary: calculates count, mean, and standard deviation of social-media use by Education_Level.

Descriptive statistics

The notebook calls `df.describe()` to summarize numeric columns using count, mean, standard deviation, minimum, quartiles, and maximum. This supports an initial understanding of the range and spread of each numeric variable.

Variable	Mean	Median	Std. dev.	Observed range
Age (years)	19.04	19.00	3.77	13.00–25.00
Social media (hours/day)	4.54	4.50	2.40	0.00–14.00
AI tool use (hours/day)	2.59	2.52	1.68	0.00–9.50
Sleep (hours/day)	6.55	6.54	1.27	2.00–11.15
Physical activity (hours/day)	1.25	1.13	1.04	0.00–5.00
Mental-health score	72.49	73.75	9.24	32.56–91.76
Physical-health score	88.02	89.45	9.69	48.03–99.98

Key Descriptive Results

Average daily habits and health scores

- Social-media use averages 4.54 hours per student per day.
- AI-tool use averages 2.59 hours per student per day.
- The sum of those two averages is 7.13 hours/day; the mean of the per-record combined usage measure is 7.13 hours/day.
- Sleep averages 6.55 hours/day.
- Physical activity averages 1.25 hours/day.
- The mean mental-health score is 72.49; the mean physical-health score is 88.02.

Category distributions

Classification	Category	Records	Share
Age group	13–15	3,682	23.0%
Age group	16–18	3,639	22.7%
Age group	19–21	3,652	22.8%
Age group	22–25	5,027	31.4%
Overall health category	At Risk	141	0.9%
Overall health category	Average	3,659	22.9%
Overall health category	Healthy	12,200	76.2%
Sleep category	Short (<6h)	5,364	33.5%
Sleep category	Normal (6–8h)	8,643	54.0%
Sleep category	Long (>8h)	1,993	12.5%
AI usage group	Low (0–1h)	3,221	20.1%
AI usage group	Moderate (1–3h)	6,421	40.1%
AI usage group	High (3–5h)	4,991	31.2%
AI usage group	Very High (5h+)	1,367	8.5%

Interpretation of the Visualizations

Chart 1 — Distribution of Daily Social-Media Use

A histogram with 30 bins and a kernel-density estimate shows the distribution of Daily_Social_Media_Hours. A dashed vertical line is intended to mark the mean.

Interpretation and review: Use it to see common usage ranges, spread, skew, and possible extreme values. The notebook's label string uses `{x.mean():.2f}` without an f-string, so the displayed legend may show the literal expression rather than the calculated mean. The plotting code should use an f-string or a preformatted label.

Chart 2 — Student Health-Wise Distribution

A pie chart shows the counts and percentages of records in Health_Category.

Interpretation and review: It summarizes the category mix after Overall_Health_Score is calculated. The notebook has `plt.title=('Student Healh Wise Distribution')`, which assigns a string to the function name instead of calling it; the intended title may therefore not display. Use `plt.title('Student Health-Wise Distribution')`.

Chart 3 — Students by Age Group

A count plot displays the number of records in each Age_Group and adds bar labels.

Interpretation and review: It describes sample composition across the four age bands. It does not by itself compare health outcomes across ages.

Chart 4 — Students by Education Level

A count plot displays record counts by Education_Level, with rotated category labels and count labels.

Interpretation and review: It helps identify which education levels are more represented in this dataset. Differences in sample counts should be considered when comparing group means.

Chart 5 — Social-Media Use Across Education Levels

A box plot compares Daily_Social_Media_Hours across education levels. A summary table below it provides count, mean, and standard deviation.

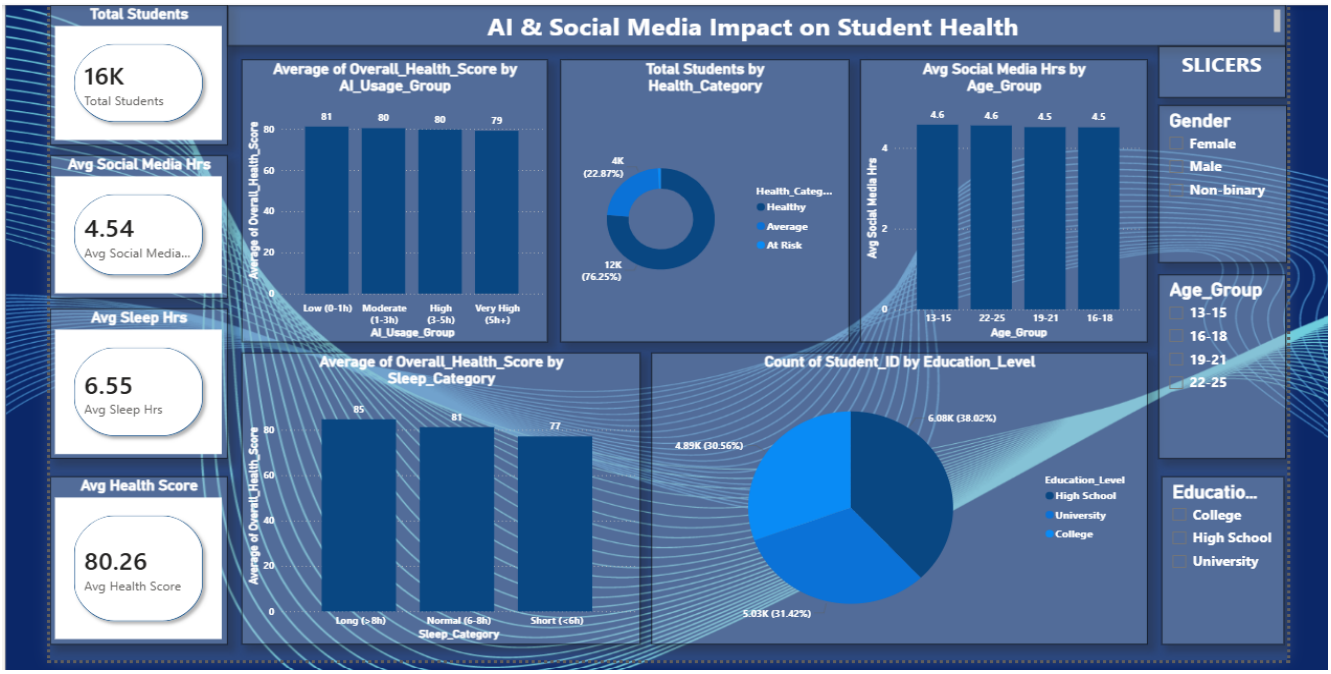
Interpretation and review: The median line, interquartile range, whiskers, and potential outliers show differences in distribution. Actual group conclusions should be based on the chart and summary values, not assumed in advance.

Chart 6 — Daily Social-Media Use vs Mental-Health Score

A scatter plot places daily social-media hours on the x-axis and Mental_Health_Score on the y-axis, with points colored by Gender.

Interpretation and review: It can reveal a possible positive, negative, or unclear visual association and whether patterns appear to differ by gender. A scatter plot alone cannot establish causation.

Power BI Dashboard



Based on the dashboard created from our dataset of 16,000 students, here is the comprehensive data story summarizing how digital habits influence student well-being.

- 1. Executive Summary & Key KPIs Our dashboard tracks the lifestyle habits and health outcomes of 16,000 students surveyed between 2025 and 2026. The high-level indicators reveal some standard benchmarks:
- Total Sample Size: 16K students.
- Average Social Media Usage: 4.54 hours per day.
- Average Sleep: 6.55 hours per night (which falls slightly below the recommended 7–8 hours for youth).
- Average Overall Health Score: 80.26 out of 100 (a solid baseline, but with room for improvement).

Key Findings & Core Insights

- A. AI Usage vs. Overall Health (The Tech-Utility Balance)
- What the dashboard shows:
- The chart Average of Overall_Health_Score by AI_Usage_Group reveals that health scores remain remarkably stable across different AI usage tiers. High-tech users (5h+ daily) maintain an average health score of 79, compared to 81 for low users (0-1h).
- The Story:
- Unlike social media, daily usage of AI tools does not show a sharp negative correlation with student health. This suggests that students may be utilizing AI productively (e.g., for education and task automation) without experiencing severe lifestyle disruption.
- B. The Critical Role of Sleep in Overall Health What the dashboard shows: In the Average of Overall_Health_Score by Sleep_Category chart, there is a clear, step-down trend: Long Sleepers (>8h) boast the highest average health score of 85. Normal Sleepers (6-8h) have an average health score of 81. Short Sleepers (<6h) experience a noticeable drop to 77. The Story: Sleep is the single most influential biological lever in this dataset. Restricting sleep to under 6 hours correlates directly with a decline in mental and physical health scores.
- C. Consistent Social Media Habits Across Ages What the dashboard shows: The Avg Social Media Hrs by Age_Group chart shows practically identical behavior across all age cohorts: 13-

15 years: 4.5 hours/day 16-18 years: 4.5 hours/day 19-21 years: 4.5 hours/day 22-25 years: 4.6 hours/day

- The Story:
- Social media engagement is deeply normalized and completely uniform across a student's academic journey—from early high school through university graduation.
- D. Health Category Breakdown
- What the dashboard shows: The Total Students by Health_Category donut chart reveals: 76.25% (12K) of students are classified as Healthy. 22.87% (4K) are classified as Average. A small portion is At Risk.
- The Story:
- While more than three-quarters of the student population is currently in a healthy state, nearly a quarter are sliding into "Average" or "At Risk" categories, requiring early intervention before lifestyle patterns solidify.
- E. Demographic Composition
- What the dashboard shows:
- The academic distribution in the Count of Student_ID by Education_Level pie chart shows: High School: 6.08K (38.02%) University: 5.03K (31.42%) College: 4.89K (30.56%) The Story: The sample is well-balanced across education tiers, lending high credibility to the behavioral patterns observed.
- 3. Strategic Recommendations
- Prioritize Sleep Campaigns: Schools and universities should focus on "sleep hygiene" rather than just screen-time limits, as sleep deprivation has the strongest correlation with lower overall health. Healthy Social Media Guardrails: Since social media usage is consistently high (4.5+ hours/day) across all age groups, institutions should promote active, creative, or community-building digital habits rather than passive, endless scrollin.Encourage Responsible AI Productivity: Because AI tool usage does not severely harm health, educators should continue integrating AI tools into studies safely, showing students how to save time and reduce academic stress.

Conclusion:

This project turns a 16,000-record student dataset into a concise dashboard and exploratory report. The strongest descriptive signals are the average social-media use of 4.54 hours/day, the negative association between social-media hours and mental-health scores, and positive associations between sleep/physical activity and mental-health scores. These are associations within the supplied dataset, not causal proof. The dashboard is most useful as a starting point for discussion and further analysis. Before using it for formal policy or high-impact decisions, verify the data source, document how scores were created, conduct appropriate statistical checks, and evaluate interventions with reliable outcome measures.